

Quantum Computing Exam

23 June 2025

Problem 1. Consider the quantum circuit corresponding to the operator $\mathbf{C}_{21}\mathbf{H}_2\mathbf{X}_2$ (the indices refer to the qubit of the register to which they are applied, assuming the enumeration 2, 1, 0 from left to right).

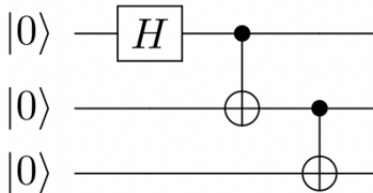
- Apply the circuit to the input register $\frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)$ obtaining the state $|\psi\rangle$.
- Measure the first qubit of $|\psi\rangle$ in the computational state and compute the probability of obtaining 1 and the state after the measurement.
- Measure the first two qubits of $|\psi\rangle$ in the Bell state and compute the probability of obtaining 0 and the state after the measurement.

Problem 2 (alternative to Problem 3). Alice wants to send to Bob a qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. To do so, Alice uses the teleportation protocol.

Compute the state of the register at each stage of the quantum circuit for this protocol assuming that Alice and Bob can share the entangled pair

$$\frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle.$$

Problem 3 (alternative to Problem 2). Work out the output of the circuit below, and show that it is an entangled state.



Problem 4 (alternative to Problem 5). Show a quantum circuit which realises the operation $C^5(U)$, where U is a unitary matrix on 1 qubit by using only CNOT and operations on 1 qubit.

Problem 5 (alternative to Problem 4). Construct a quantum circuit implementing the two-level matrix

$$U = \begin{bmatrix} a & 0 & 0 & 0 & 0 & 0 & 0 & c \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ b & 0 & 0 & 0 & 0 & 0 & 0 & d \end{bmatrix}$$

by using only $\mathbf{C}^2(\mathbf{NOT})$ and $\mathbf{C}^2(\mathbf{A})$, where $\mathbf{A} = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$. Write down the matrix corresponding to the $\mathbf{C}^2(\mathbf{NOT})$ gates of the resulting circuit.

Problem 6 (optional). Assume that it is given a circuit implementing the operator $\mathbf{U}_f$ for one of the following four functions $f_i(x): \{0, 1\} \rightarrow \{0, 1\}$ defined in the table below. The circuit is a black box, so we do not know which function is actually implemented by $\mathbf{U}_f$.

i	$x = 0$	$x = 1$
f_0	0	0
f_1	0	1
f_2	1	0
f_3	1	1

- Describe an algorithm which determines whether the function f belongs to $\{f_0, f_3\}$ or to $\{f_1, f_2\}$ by making the minimum number of queries to the oracle $\mathbf{U}_f$.